

Beyond Semantic Confidence: Relation-Consistent Geometric Re-ranking for 3D Scene Graphs

Supplementary Material

Anonymous submission

Model and Preprocessing Details

Open3DSG. Open3DSG uses the official non-averaged BLIP checkpoint `epoch=13-step=13104.ckpt`, selected by train/dev validation loss. Public preprocessing produces candidates for 533/548 contexts; the main evaluation retains the full 548-context ground-truth universe and treats the 15 contexts without candidates as empty sets. For sensitivity, the eligible-only denominator contains 3,899 relations, and a documented recovery covers all 548 contexts by lowering the visible-object minimum to two and regenerating views for two scans.

SGFN. We use the released `SGFN_full_1160` model from the SceneGraphFusion 3DSSG implementation, with its 160-object, 26-relation checkpoint on the same evaluation scans and denominator.

Guarantees and Compatibility Analyses

Proposition 1 (Exact transformation consistency) *Let a a finite transformation group H_a act jointly on the predicate and geometry of an ordered pair, and define $C_a^{\text{tr}}(x) = |H_a|^{-1} \sum_{g \in H_a} C_a(gx)$. Then $C_a^{\text{tr}}(hx) = C_a^{\text{tr}}(x)$ for every $h \in H_a$.*

Proof. For any $h \in H_a$, $C_a^{\text{tr}}(hx) = |H_a|^{-1} \sum_{g \in H_a} C_a(ghx)$. Right multiplication by h is a bijection on the finite group, so re-indexing $g' = gh$ gives $C_a^{\text{tr}}(hx) = C_a^{\text{tr}}(x)$. The two-element proximity and vertical actions yield the identities in the main paper, and the singleton support/contact action is the identity.

Family-sequence preservation. For each family a , the algorithm constructs the ordered list π_a defined in the main paper. At position j , it selects the next unselected candidate from π_{a_j} ; hence the output family sequence is exactly $(a_1, \dots, a_n)$. The support/contact list follows the source ranking, so the first K positions contain exactly the same ordered support/contact subsequence as the source prefix. Therefore the support/contact selections and their order are identical at every reported K .

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Prefix utility optimality. Fix a prefix length K , and let $n_a(K)$ be the number of positions assigned to family a in the source prefix. Every feasible ranking that preserves these family counts must select $n_a(K)$ candidates from each family. Because each re-ranked π_a is sorted by its variant’s u_i , either RelCompat3D estimator selects the $n_a(K)$ highest-utility candidates within that family. If another feasible selection omits one of these candidates in favor of a lower-utility candidate from the same family, exchanging the two increases the summed within-prefix utility. The support/contact subsequence is fixed, so it does not affect this comparison.

Counterfactual target construction. Measured-family ground-truth relations provide positives. Negatives are generated within the same scene context and linked to their source positive; no predictor score, predictor identity, or evaluation candidate is used. Table 1 gives the complete construction policy.

At most two negatives are emitted per positive, 200 per context–family, and three times the number of positives per family. These caps limit repeated easy negatives; they do not equalize family counts. The export contains 28,977 positives and 37,477 negatives. Total rows by family are 23,008 proximity, 5,488 vertical, and 37,958 support/contact. The explicit overlap above is why Violation computed by the rule-based geometry verifier is reported as a verifier-derived measure rather than independent physical-validity ground truth; the feature-removal analysis tests, but does not eliminate, this construct dependence.

RelCompat3D-Linear fits one logistic model per measured relation family. Numeric features use means and standard deviations estimated from the training split; missing values use the corresponding training mean. Inputs are predicate indicators T , predicate-independent geometric measurements of the ordered pair G , and the two predicate-signed height interactions defined in the main paper. The family label selects the head and is not repeated as an input. The predictor score, source rank, predictor identity, and object class are excluded. Transformation-consistent augmentation preserves the proximity predicate under endpoint swap; vertical order swaps endpoints and applies the inverse predicate. Linked counterfactuals receive the fixed margin loss in the main paper. Support/contact receives the pairwise loss but no blanket endpoint transform.

Family	Intervention and acceptance rule	Relation to verifier
Proximity	Choose directed same-context pairs in descending normalized XY distance; require distance ≥ 2.5 and zero projected overlap.	Shares distance/overlap primitives and the 2.5 satisfied boundary; the verifier declares violation only at ≥ 3.5 .
Vertical	On the same directed pair, replace higher than with lower than or conversely; require the positive direction to have $ \Delta z \geq 0.25$ m and normalized $ \Delta z \geq 0.15$.	Shares both signed primitives and thresholds.
Support/contact	Replace the subject or object within context; reject known support/contact pairs in either direction and floor subjects; require any two of normalized XY distance ≥ 2.0 , zero overlap, and vertical gap ≥ 0.30 m.	Shares these OBB primitives and thresholds; point-subtype evidence used by the reported verifier is not used to generate negatives.

Table 1: Counterfactual-negative construction. Normalized XY distance uses the mean object-pair XY diagonal. Zero overlap requires both directed projected-overlap ratios to be at most 10^{-9} .

RelCompat3D-MLP fits a single shared two-hidden-unit ReLU model with a predicate-linear skip connection. Its 29 inputs comprise family and predicate indicators, the same 17 geometric measurements, the two signed-height interactions, and the sum of the directional overlap ratios. The family indicator is nonconstant because this estimator shares parameters across families. It has 69 parameters and excludes the predictor score and identity.

The 60,208 training rows yield 33,961 linked pairs, and transformation-consistent augmentation produces 86,032 optimization rows for both estimators. Linear uses 800 deterministic full-batch steps at learning rate 0.2; MLP uses 120 steps at learning rate 0.02. Both use the same pairwise weight, margin, ℓ_2 coefficient, and fixed seed policy. The three linear family models contain 21–23 parameters. The support/contact head is retained for the Product (all families) comparison, while neither proposed ranking uses support/contact compatibility and both preserve that family’s source ordering.

Computational cost. We benchmark the fixed RelCompat3D-Linear CPU implementation with one process and one thread on an Intel Core Ultra 7 265KF. Timing begins from preloaded relation candidates with pre-computed geometric measurements of the ordered pair and includes proximity/vertical compatibility, transformation averaging, family-specific list sorting, and output assembly. It excludes source-predictor inference, reconstruction and geometry association, file parsing, metrics, and bootstrap, so

Predictor	Contexts	Scored candidates	Total (s)	ms/context
VL-SAT	548	110,424	2.445	4.461
Open3DSG	533	79,722	1.808	3.391
SGFN	548	110,424	2.453	4.476

Table 2: CPU re-ranking cost for RelCompat3D-Linear. Open3DSG has 533 nonempty public-prediction contexts; its 15 empty official contexts require no scoring or sorting.

Family	Train	Dev	Params	Pairs	AUROC	Brier
Proximity	20,902	2,106	21	9,562	1.0000	.0009
Vertical order	4,922	566	22	1,644	.9982	.0079
Support/contact	34,384	3,574	23	22,755	.9933	.0256

Table 3: RelCompat3D-Linear training and development discrimination. Labels comprise GT-positive/counterfactual pairs, not human physical-validity judgments.

it is not end-to-end pipeline latency. Table 2 reports medians over five runs after one warm-up.

The three stored linear heads contain 66 parameters; the primary proximity/vertical path evaluates 43 of them and adds no fitted fusion parameter. Peak process RSS, including the preloaded candidates, is 366.5 MiB. All five runs preserve the exact family sequence and support/contact order.

Across all 6,246 development rows, AUROC/AUPRC/Brier are .9973/.9973/.0157. On 3,516 linked development pairs, the positive wins 99.23% of the time with mean logit margin 14.14. Correct-predicate compatibility wins on 97.40% of 384 vertical positives. Exact transformation averaging gives a maximum compatibility difference of zero under proximity endpoint swap and under joint endpoint swap and inverse-predicate transformation for vertical order, on development and evaluation rows from all three predictors. The shared nonlinear estimator obtains development AUROC/AUPRC/Brier of .9904/.9926/.0225. Its transformation-averaged compatibility also changes monotonically in all 512 proximity and 512 vertical intervention cases used by the surface audit.

The factor ablation uses the same split and estimator class in a separate pooled analysis; because one head spans all families there, its semantic input includes predicate and family identifiers. At development, semantic-only, predicate-independent G -only, additive semantics+ G , and interaction $T \times G$ obtain AUROC .6124/.8809/.9193/.9822 and Brier .2370/.1363/.1061/.0495. Shuffled and mismatched-pair geometry reduce the intended reliability signal. These results verify factor use and the implemented transformation identities; they do not convert the constructed target into independent physical-validity ground truth.

Every feature-removal model is refitted on the 1,061 training scans with the same linked positive-counterfactual objective, exclusion of the predictor score, and exact transformation averaging. The exact-scalar condition retains development AUROC of .99999 for proximity and .99794 for vertical order. Removing all related distance or vertical measurements lowers these to .9512 and .7089; alternative ev-

Source	Condition	$K = 50$	$K = 100$
VL-SAT	Source score	.9272/.0268	.9635/.0476
	Full model	.9277/.0197	.9658/.0295
	Exact verifier scalar removed	.9280/.0199	.9660/.0300
	Related measurements removed	.9275/.0268	.9653/.0468
	Alternative evidence only	.9280/.0268	.9642/.0471
Open3DSG	Source score	.4043/.1387	.5111/.1242
	Full model	.4418/.0342	.5685/.0324
	Exact verifier scalar removed	.4381/.0342	.5687/.0325
	Related measurements removed	.4119/.0954	.5264/.0955
	Alternative evidence only	.4104/.1194	.5327/.1188
SGFN	Source score	.7402/.0385	.9235/.0630
	Full model	.7450/.0263	.9303/.0350
	Exact verifier scalar removed	.7465/.0268	.9300/.0358
	Related measurements removed	.7467/.0388	.9298/.0616
	Alternative evidence only	.7462/.0386	.9277/.0625

Table 4: Feature-removal analysis. Cells are Recall / verifier V under the same family-aware ranking procedure. Exact removes the normalized scalar consumed by the verifier; family removes all directly related distance or vertical measurements; alternative retains only projected overlap for proximity and horizontal-distance/overlap context for vertical order.

idence yields .8763 and .5531. At $K = 50$, the strict conditions retain a clear Open3DSG improvement but not the full V gain on VL-SAT or SGFN. At $K = 100$, all feature-removal conditions improve both point metrics across all three predictors. The results show that performance does not depend solely on a single verifier input, but they do not eliminate broader construct dependence on correlated geometry. The main paper reports the matched controls under the active family-aware ranking procedure; the separate all-family product and pooled comparison appears in Table 12.

Surface-Based Construct-Validity Audit

The primary audit covers the proximity and vertical families re-ranked by RelCompat3D. It does not assess support/contact, whose candidates remain in source order and require a separate contact construct. Point measurements use instance vertices: a symmetric 10th-percentile nearest-surface distance normalized by a robust point radius for proximity, and the difference between median vertical coordinates for vertical order. Mesh measurements use area-weighted triangle centroids from the reconstructed instance surfaces and an area-derived scale. They do not read OBB measurements, predictor identity, predictor scores, compatibility scores, or primary verifier labels.

All thresholds are fixed using positive relations from the 1,061-scan training split. Point measurements require 49 samples per endpoint and use normalized proximity

boundaries 0.6217/1.1645 and a 0.0941 m absolute vertical margin; mesh measurements require 56 samples and use 0.7230/1.4521 and 0.0826 m. The lower and upper proximity boundaries are the training-positive 90th and 99th percentiles, and the vertical margin is the 10th percentile of the absolute signed separation. A strict consensus label is satisfied or violated only when point and mesh agree; measured disagreements are uncertain. Unsupported rows do not enter the audit-V denominator.

The audit has raw-surface data for all 157 evaluation scans. At $K = 50/100$, consensus measured/decidable coverage for Linear is 95.5/71.0 and 95.2/73.8% on VL-SAT, 98.2/83.7 and 98.1/82.5% on Open3DSG, and 95.8/79.9 and 95.8/77.6% on SGFN. The corresponding MLP values are 95.6/71.3 and 95.4/74.1%, 98.3/85.7 and 98.1/82.9%, and 95.8/80.2 and 95.9/78.1%. Measured disagreements remain uncertain and enter the denominator but not the violation numerator.

As a mechanism test, 512 synthetic interventions per family translate one endpoint away from the other for proximity or across the vertical decision boundary. Both compatibility estimators respond monotonically in all proximity and vertical cases. The point and mesh measurements respond monotonically in 96.29/94.73% of proximity cases and 100% of vertical cases; surface quantiles can be non-monotone under translation for non-convex objects. The audit therefore reduces dependence on the exact OBB rules used in the primary metric, but both surface estimates still use the same reconstructed 3RScan geometry and relation ontology. They are not independent physical-validity ground truth.

Qualitative Pair Analysis

Figure 1 links each ordered-pair projection to the measured evidence and ranking outcome. The proximity and vertical examples are demoted by both capacity variants. The support/contact example is retained in source order because that family is outside the re-ranking scope.

Counterfactual-Policy Sensitivity

We regenerate the training and internal-development targets and refit the proximity and vertical heads under nine one-factor-at-a-time conditions. Each variant recomputes normalization from the 1,061 training scans; no final-validation row enters target generation, normalization, or fitting. The default refit reproduces the main model and metrics exactly. Support/contact is kept in source order in every condition.

Across all variants, the maximum absolute change from the default is .0020 Recall and .0011 V at $K = 50$, and .0035 Recall and .0020 V at $K = 100$. Every variant retains lower V and no lower Recall than source scoring for all three predictors at $K = 50$ and $K = 100$. This reduces dependence on a particular threshold or negative count, but does not make the constructed target independent of the verifier.

Capacity, Fusion, and Open3DSG Sensitivities

Compatibility capacity and matched fusion. RelCompat3D-Linear uses family-specific linear heads, whereas RelCompat3D-MLP uses a shared two-hidden-unit

Predictor	K	Point			Mesh			Consensus		
		Source	Linear	MLP	Source	Linear	MLP	Source	Linear	MLP
VL-SAT	5	1.36	0.93	0.85	2.19	1.43	1.18	0.93	0.51	0.42
	10	5.70	5.11	4.67	5.30	4.65	4.31	3.61	3.06	2.79
	20	10.45	9.47	9.25	9.46	8.55	8.27	7.27	6.36	6.17
	50	20.11	18.11	17.74	19.50	17.51	17.19	16.43	14.34	14.05
	100	29.28	24.89	24.90	29.44	25.09	25.12	26.00	21.50	21.57
Open3DSG	5	83.18	1.48	17.16	83.89	1.62	17.52	83.20	1.26	16.84
	10	65.96	2.18	13.64	65.94	2.10	13.83	65.18	1.60	13.18
	20	55.29	3.89	9.80	55.07	3.93	10.07	53.80	2.90	9.17
	50	47.76	6.28	10.02	47.85	6.57	10.27	45.96	4.90	8.84
	100	42.65	10.25	13.99	43.05	10.77	14.38	40.87	8.65	12.32
SGFN	5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	10	3.06	2.18	1.53	3.02	1.94	1.72	2.42	1.54	1.32
	20	6.68	5.17	4.61	6.90	5.44	4.92	5.33	3.89	3.55
	50	14.05	10.73	10.09	13.61	10.26	9.71	11.56	8.13	7.63
	100	24.45	19.05	18.89	24.33	18.96	18.75	21.55	15.98	15.88

Table 5: Surface-based Violation percentages across all reported budgets. Point and mesh use distinct surface summaries; Consensus requires their binary labels to agree. This metric is a separate construct and is not directly comparable to primary Violation.

ReLU model with a predicate-linear skip connection. They share the constructed training target, linked-pair objective, transformation averaging, exclusion of predictor scores and identities, product utility, and family-aware ranking. The MLP input includes a family indicator because its parameters are shared, plus an overlap-sum interaction derived from the same directional overlap measurements. Both are fitted once and applied to all three predictors without target-specific refitting. Rank-average fusion (RankAvg) and reciprocal rank fusion (RRF) use the linear compatibility scores and the same family-aware ranking procedure.

Let q_i^Z and q_i^C denote descending within-context percentile ranks of the predictor score and compatibility, and let r_i^Z, r_i^C be their one-indexed ranks. Rank-average and fixed-constant Reciprocal Rank Fusion use

$$S_i^{\text{rank}} = \frac{1}{2}(q_i^Z + q_i^C), \quad S_i^{\text{RRF}} = \frac{1}{60 + r_i^Z} + \frac{1}{60 + r_i^C}. \quad (1)$$

At $K = 50$, the nonlinear variant has higher Open3DSG Recall than the linear variant but also higher V; they are nearly tied on VL-SAT and SGFN. Rank fusion can lower V at large budgets but incurs larger predictor-dependent Recall losses. No estimator or fusion dominates the joint trade-off. A separate exact-match MLP uses stronger SGFN-specific correctness supervision and is therefore retained only as a source-specific upper comparator, not as an apples-to-apples baseline.

Matched structural controls. Across both budgets and model capacities, corrupted predicates and pair geometry increase verifier-derived Violation, while compatibility-only ordering loses substantial Recall on VL-SAT and SGFN. Open3DSG remains more sensitive to predictor-score scale.

Pooled compatibility ablation. The pooled product fits one compatibility model across all families and does not use

the primary family-aware ranking procedure. Compared with Product (all families), it can increase Recall but generally raises V, particularly for Open3DSG and SGFN:

Open3DSG coverage sensitivity. The public preprocessing drops a context when fewer than four annotated objects retain view metadata. This accounts for all 15 missing contexts. Table 13 distinguishes the eligible-context evaluation, the conservative full-target main evaluation, and a documented full-coverage recovery.

The conclusion is stable to all three denominator and coverage choices. The main paper uses the conservative full-target public evaluation; recovered results are sensitivity evidence rather than the headline benchmark.

Scan-Level Cluster Bootstrap

The analysis resamples the 157 scan identifiers 1,000 times and carries every context of a sampled scan together. The ranking, model, and point estimates remain unchanged; shared bootstrap indices are used across conditions. A paired 548-context bootstrap gives the same directions and is retained as a finer-grained sensitivity.

For both capacities, all Recall point changes are non-negative. Open3DSG Recall intervals are strictly positive at every K , and the SGFN $K = 50/100$ intervals are positive; VL-SAT support depends on capacity and budget. Every Violation interval is strictly below zero except the tied SGFN $K = 5$ point and its $K = 10$ interval, which reaches zero. Thus the point-estimate trend is broad, while statistical support remains predictor- and budget-dependent.

External Transfer Stress Test

We apply the 3DSSG-trained compatibility model and family-aware ranking rule to the official ReplicaSSG test

Predictor	K	Point ΔV (95% CI)	Mesh ΔV (95% CI)	Consensus ΔV (95% CI)
VL-SAT	5	-0.42 [-0.83, -0.09]	-0.76 [-1.37, -0.26]	-0.42 [-0.83, -0.09]
	10	-0.59 [-0.99, -0.26]	-0.65 [-1.05, -0.34]	-0.55 [-0.94, -0.25]
	20	-0.99 [-1.30, -0.71]	-0.91 [-1.21, -0.65]	-0.91 [-1.22, -0.64]
	50	-2.00 [-2.30, -1.72]	-1.99 [-2.28, -1.73]	-2.09 [-2.41, -1.80]
	100	-4.40 [-4.80, -4.06]	-4.35 [-4.73, -4.02]	-4.51 [-4.90, -4.17]
Open3DSG	5	-81.70 [-83.66, -79.57]	-82.26 [-84.29, -80.15]	-81.95 [-83.90, -79.84]
	10	-63.78 [-65.82, -61.58]	-63.84 [-65.87, -61.64]	-63.58 [-65.68, -61.33]
	20	-51.40 [-53.18, -49.55]	-51.14 [-52.95, -49.31]	-50.90 [-52.68, -49.01]
	50	-41.48 [-43.11, -39.73]	-41.28 [-42.93, -39.53]	-41.06 [-42.75, -39.28]
	100	-32.39 [-33.82, -30.88]	-32.28 [-33.66, -30.86]	-32.22 [-33.61, -30.74]
SGFN	5	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
	10	-0.87 [-2.09, 0.23]	-1.08 [-2.24, -0.20]	-0.88 [-1.94, -0.00]
	20	-1.51 [-2.26, -0.89]	-1.46 [-2.19, -0.88]	-1.44 [-2.22, -0.84]
	50	-3.32 [-3.80, -2.90]	-3.35 [-3.83, -2.95]	-3.43 [-3.92, -3.00]
	100	-5.39 [-5.83, -4.96]	-5.37 [-5.79, -4.95]	-5.57 [-6.03, -5.12]

Table 6: Paired surface-audit changes for RelCompat3D-Linear. Each entry is Linear minus Source surface-based Violation in percentage points with a paired scan-level cluster 95% interval. The SGFN result at $K = 5$ is tied; the point-only SGFN interval at $K = 10$ contains zero.

scenes using FROSS predictions. No target-specific parameter fitting, normalization, or imputation is performed in this evaluation. The exact mapping covers `near`, `above`, and `under`; `support/contact` is excluded because the ontologies do not provide an exact mapping. We report this as a transfer stress test rather than evidence of dataset-level generalization.

Paired scene-bootstrap intervals support joint improvement at $K = 10$ (ΔR [.0048, .1467], ΔV [-0.1000, -0.0091]) and $K = 50$ (ΔR [.0179, .0811], ΔV [-0.0683, -0.0185]). The $K = 20$ Recall interval touches zero, while $K = 5$ is inconclusive. At $K = 100$, heavy quantization of the predictor score makes multiplicative re-ranking nearly inert, and candidate support caps exact-match Recall at 44.19%. The stress test therefore supports budget-dependent transfer, not dataset-level generalization.

Family Composition within the Selected Top-100 Predictions

Table 17 reports each family’s slice inside the actual global top-100 ranking. Deltas compare RelCompat3D with the predictor score using shared paired context-bootstrap indices.

Support/contact is exactly unchanged because the ranking procedure preserves both its selection and the source family sequence at every K . Vertical order drives most of the aggregate reduction. The result is consequently a scoped operating-point improvement, not a uniform family claim. The same direction holds across the reported budgets; full machine-readable family slices and simultaneous intervals accompany the code artifact.

Verifier-Uncertainty Sensitivity

The reported $V@K_{all}$ includes uncertain candidates in its denominator but does not label them satisfied. We therefore recompute each ranking with $V@K_{dec}$ over satisfied/violated candidates, uncertainty rate $U@K$, and a pessimistic bound

$V@K_{pess}$ that counts every uncertain candidate as a violation. Table 18 gives the $K = 100$ comparison.

RelCompat3D-minus-source changes in V_{dec} are $-.0277$ for VL-SAT, $-.1578$ for Open3DSG, and $-.0440$ for SGFN; their paired 95% intervals all exclude zero. Uncertainty-rate changes are $+.0009$, $-.0076$, and $+.0071$, whereas pessimistic- V changes are $-.0173$, $-.0993$, and $-.0209$. Thus both decidable-subset and pessimistic V are lower for all three predictors even though uncertainty does not decrease uniformly. The hard-rule baseline has zero Violation when the denominator includes all verifier statuses, but it retains uncertain candidates and therefore has pessimistic V of $.3712/.4722/.4042$. Thus zero measured violation does not imply reliable or complete selection.

Predictor	K	Point ΔV (95% CI)	Mesh ΔV (95% CI)	Consensus ΔV (95% CI)
VL-SAT	5	-0.51 [-0.94, -0.17]	-1.01 [-1.85, -0.34]	-0.51 [-0.95, -0.17]
	10	-1.03 [-1.50, -0.64]	-0.99 [-1.47, -0.61]	-0.83 [-1.27, -0.46]
	20	-1.21 [-1.56, -0.89]	-1.19 [-1.54, -0.86]	-1.10 [-1.43, -0.78]
	50	-2.37 [-2.71, -2.06]	-2.31 [-2.65, -2.01]	-2.38 [-2.73, -2.07]
	100	-4.39 [-4.79, -4.05]	-4.33 [-4.71, -4.02]	-4.43 [-4.84, -4.10]
Open3DSG	5	-66.02 [-68.35, -63.30]	-66.37 [-68.53, -63.80]	-66.36 [-68.57, -63.78]
	10	-52.32 [-54.26, -50.28]	-52.11 [-54.11, -50.07]	-52.01 [-54.05, -49.98]
	20	-45.50 [-47.35, -43.60]	-45.00 [-46.74, -43.18]	-44.63 [-46.41, -42.78]
	50	-37.74 [-39.47, -35.99]	-37.58 [-39.38, -35.78]	-37.12 [-38.96, -35.28]
	100	-28.66 [-30.00, -27.26]	-28.67 [-30.07, -27.26]	-28.56 [-29.96, -27.12]
SGFN	5	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
	10	-1.53 [-2.85, -0.46]	-1.30 [-2.54, -0.39]	-1.11 [-2.27, -0.21]
	20	-2.07 [-2.86, -1.36]	-1.98 [-2.73, -1.36]	-1.78 [-2.54, -1.14]
	50	-3.96 [-4.50, -3.45]	-3.90 [-4.47, -3.39]	-3.93 [-4.48, -3.43]
	100	-5.56 [-6.00, -5.11]	-5.58 [-6.03, -5.13]	-5.67 [-6.15, -5.22]

Table 7: Paired surface-audit changes for RelCompat3D-MLP. Each entry is MLP minus Source surface-based Violation in percentage points with a paired scan-level cluster 95% interval. The SGFN result at $K = 5$ is tied.

(a) $K = 50$				
Condition	Order	VL-SAT	Open3DSG	SGFN
Default	1.000	.9277/.0197	.4418/.0342	.7450/.0263
Proximity 2.0	1.000	.9280/.0197	.4436/.0342	.7470/.0263
Proximity 3.0	1.000	.9275/.0198	.4426/.0342	.7437/.0264
Vertical 0.20 m	1.000	.9277/.0194	.4411/.0342	.7450/.0258
Vertical 0.30 m	1.000	.9277/.0204	.4439/.0342	.7447/.0274
Negative cap 1	1.000	.9277/.0197	.4428/.0342	.7447/.0263
Negative cap 4	1.000	.9277/.0197	.4416/.0342	.7455/.0263
Pair weight .125	1.000	.9277/.0198	.4421/.0342	.7450/.0264
Pair weight .5	1.000	.9277/.0197	.4421/.0342	.7450/.0262
(b) $K = 100$				
Condition	Order	VL-SAT	Open3DSG	SGFN
Default	1.000	.9658/.0295	.5685/.0324	.9303/.0350
Proximity 2.0	1.000	.9660/.0294	.5720/.0324	.9325/.0350
Proximity 3.0	1.000	.9653/.0296	.5662/.0324	.9277/.0350
Vertical 0.20 m	1.000	.9658/.0289	.5675/.0324	.9308/.0334
Vertical 0.30 m	1.000	.9658/.0306	.5672/.0324	.9300/.0370
Negative cap 1	1.000	.9658/.0295	.5700/.0324	.9293/.0350
Negative cap 4	1.000	.9658/.0295	.5670/.0324	.9305/.0350
Pair weight .125	1.000	.9658/.0296	.5690/.0324	.9303/.0351
Pair weight .5	1.000	.9658/.0294	.5682/.0324	.9303/.0347

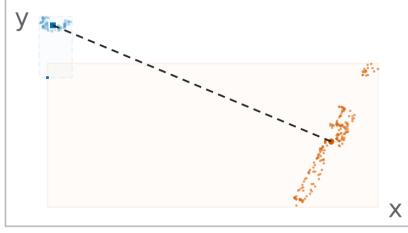
Table 8: Counterfactual-policy sensitivity. Order is linked-positive ordering accuracy over the re-ranked proximity and vertical sensitivity pairs on the 117-scan internal-development split. Source cells are Recall / verifier V. Only the named factor changes from the default.

Source	Fusion	$K = 50$	$K = 100$
VL-SAT	Linear	.9277/.0197	.9658/.0295
	MLP	.9272/.0189	.9650/.0296
	RankAvg	.9023/.0162	.9705/.0226
	RRF	.9172/.0182	.9705/.0250
Open3DSG	Linear	.4418/.0342	.5685/.0324
	MLP	.4670/.0413	.5989/.0371
	RankAvg	.4320/.0460	.5400/.0549
	RRF	.4245/.0960	.5468/.0780
SGFN	Linear	.7450/.0263	.9303/.0350
	MLP	.7457/.0258	.9288/.0350
	RankAvg	.7022/.0235	.9272/.0260
	RRF	.6850/.0254	.8799/.0307

Table 9: Compatibility-capacity and fusion comparisons. Cells are Recall / verifier V at $K = 50$ and $K = 100$. Every row preserves the source family sequence and support/contact selections.

(a) Proximity correction

heater / close by / trash can

**Measured evidence****XY center distance****4.33 m**

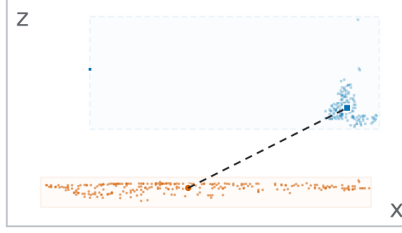
large separation for close by

Rank 19 → 178

Z=0.853; Ctr=0.003

Demoted: proximity**(b) Vertical correction**

floor / higher than / curtain

**Measured evidence****subject–object center Δz** **−1.02 m**

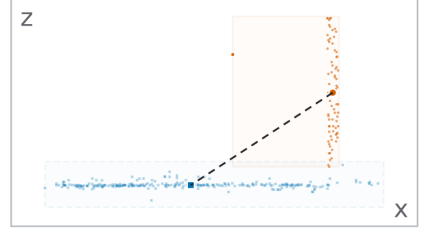
subject lies below the object

Rank 1 → 430

Z=0.871; Ctr=0.000

Demoted: vertical order**(c) Contact residual**

door / lying on / floor

**Measured evidence****vertical bottom–top gap****−0.06 m**

contact remains unresolved

Rank 21 → 21

Z=0.843; source order

Kept in source order

Figure 1: Pair–evidence–outcome analysis. Orange circles and solid boxes identify the subject; blue squares and dashed boxes identify the object. The first two violations are demoted; the residual case requiring contact evidence is kept in source order.

Condition	Open3DSG		VL-SAT		SGFN	
	Linear	MLP	Linear	MLP	Linear	MLP
Full method	44.18/3.42	46.70/4.13	92.77/1.97	92.72/1.89	74.50/2.63	74.57/2.58
Wrong predicate	42.65/21.98	46.78/19.30	90.46/4.97	90.99/5.03	71.58/9.42	71.78/9.67
Wrong pair	38.52/8.50	38.02/9.31	90.56/2.61	91.31/2.62	71.55/3.99	72.10/3.91
Shuffled geometry	38.42/12.93	36.83/12.88	89.17/3.05	90.26/3.00	70.59/4.73	71.15/4.59
Fixed-label swap	42.67/21.98	45.52/19.13	90.46/4.97	90.99/4.99	71.58/9.42	71.85/9.65
Distance only	51.16/8.24		81.90/5.34		63.19/10.00	
Compatibility only	42.22/3.42	48.19/3.42	67.65/1.61	77.84/1.61	52.79/2.32	57.10/2.33

Table 10: Matched structural controls for both compatibility estimators at $K = 50$. Cells are Recall / Violation percentages ($\uparrow$ / $\downarrow$). All conditions use the same candidates and family-aware ranking as the main evaluation. Distance only is common to both heads.

Condition	Open3DSG		VL-SAT		SGFN	
	Linear	MLP	Linear	MLP	Linear	MLP
Full method	56.85/3.24	59.89/3.71	96.58/2.95	96.50/2.96	93.03/3.50	92.88/3.50
Wrong predicate	53.47/20.96	58.89/18.34	94.79/7.95	94.99/8.31	89.93/12.99	90.01/13.52
Wrong pair	49.32/8.36	49.90/8.90	94.96/4.73	95.62/4.71	88.07/6.65	89.05/6.59
Shuffled geometry	48.79/12.42	47.94/12.38	94.11/5.60	94.71/5.49	86.83/8.36	87.84/8.06
Fixed-label swap	53.63/20.96	57.43/18.15	94.81/7.95	95.09/8.13	90.03/12.98	90.23/13.35
Distance only	63.22/9.55		89.80/8.09		84.06/12.66	
Compatibility only	56.87/3.24	61.33/3.28	83.89/2.03	90.18/2.05	70.64/2.30	80.99/2.31

Table 11: Structural controls at $K = 100$. Cells are Recall / Violation percentages ($\uparrow$ / $\downarrow$). All conditions use the same candidates and family-aware ranking as the main evaluation. Distance only is common to both heads.

Source	$K = 50$		$K = 100$	
	Family	Pooled	Family	Pooled
VL-SAT	.9293/.0203	.9300/.0219	.9688/.0325	.9690/.0387
Open3DSG	.4685/.0290	.4718/.0528	.6052/.0340	.6443/.0747
SGFN	.7704/.0258	.7925/.0292	.9413/.0371	.9413/.0464

Table 12: Family-specific and pooled products. Cells are Recall / verifier V at $K = 50$ and $K = 100$; both products apply compatibility to all relation families.

Evaluation	Contexts / GT	Source	RelCompat3D
Public eligible	533 / 3,899	.5206/.1242	.5791/.0324
Public, full target (main)	548 / 3,972	.5111/.1242	.5685/.0324
Recovered full coverage	548 / 3,972	.5161/.1242	.5735/.0332

Table 13: Open3DSG coverage sensitivity at $K = 100$. Cells are Recall / verifier V. The main evaluation uses the full 3,972-relation denominator and no predictions for the 15 public-pipeline drops.

Predictor	K	Δ Recall	95% CI	Δ Violation	95% CI
VL-SAT	5	+0.13	[0.00, 0.26]	−0.15	[−0.30, −0.03]
	10	+0.18	[0.05, 0.35]	−0.26	[−0.45, −0.09]
	20	+0.08	[−0.03, 0.19]	−0.28	[−0.43, −0.16]
	50	+0.05	[−0.05, 0.18]	−0.70	[−0.88, −0.55]
	100	+0.23	[0.03, 0.46]	−1.82	[−2.00, −1.65]
Open3DSG	5	+0.30	[0.11, 0.49]	−51.11	[−54.39, −47.86]
	10	+1.51	[1.04, 2.00]	−30.56	[−32.55, −28.51]
	20	+3.73	[2.96, 4.46]	−17.86	[−18.97, −16.74]
	50	+3.75	[2.96, 4.50]	−10.45	[−11.06, −9.86]
	100	+5.74	[4.63, 6.86]	−9.18	[−9.62, −8.69]
SGFN	5	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]
	10	0.00	[0.00, 0.00]	−0.02	[−0.06, 0.00]
	20	+0.03	[−0.07, 0.14]	−0.25	[−0.43, −0.11]
	50	+0.48	[0.29, 0.69]	−1.22	[−1.44, −1.01]
	100	+0.68	[0.42, 0.98]	−2.80	[−3.05, −2.54]

Table 14: Paired scan-level cluster intervals for RelCompat3D-Linear. Entries are Linear minus the predictor score in percentage points with 95% intervals.

Predictor	K	Δ Recall	95% CI	Δ Violation	95% CI
VL-SAT	5	+0.15	[0.00, 0.33]	−0.15	[−0.30, −0.03]
	10	+0.25	[0.07, 0.46]	−0.31	[−0.53, −0.11]
	20	+0.18	[0.05, 0.34]	−0.33	[−0.49, −0.20]
	50	0.00	[−0.10, 0.10]	−0.79	[−0.98, −0.62]
	100	+0.15	[−0.08, 0.39]	−1.80	[−1.99, −1.63]
Open3DSG	5	+0.28	[0.08, 0.48]	−47.09	[−49.94, −44.04]
	10	+1.91	[1.31, 2.48]	−27.92	[−29.77, −25.95]
	20	+4.78	[3.85, 5.72]	−16.44	[−17.51, −15.34]
	50	+6.27	[5.02, 7.49]	−9.74	[−10.31, −9.13]
	100	+8.79	[7.28, 10.23]	−8.70	[−9.13, −8.24]
SGFN	5	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]
	10	0.00	[0.00, 0.00]	−0.02	[−0.06, 0.00]
	20	+0.08	[−0.05, 0.23]	−0.26	[−0.43, −0.13]
	50	+0.55	[0.32, 0.81]	−1.27	[−1.51, −1.06]
	100	+0.53	[0.28, 0.79]	−2.79	[−3.06, −2.54]

Table 15: Paired scan-level cluster intervals for RelCompat3D-MLP. Entries are MLP minus the predictor score in percentage points with 95% intervals.

K	Source R/V	RelCompat3D R/V	Δ R/ Δ V
5	.0465/.0545	.0698/.0182	+.0233/−.0364
10	.0756/.0818	.1453/.0273	+.0698/−.0545
20	.1453/.1273	.2209/.0364	+.0756/−.0909
50	.2616/.1328	.3140/.0904	+.0523/−.0424
100	.3547/.1967	.3547/.1958	+.0000/−.0010

Table 16: ReplicaSSG/FROSS transfer across all reported budgets. R is exact-match Recall over 172 relations; V is primary Violation. The model and ranking rule match the main experiment.

Source	Family	Δ R	Δ V
VL-SAT	Support/contact	.0000	.0000
VL-SAT	Proximity	+.0040	−.0076
VL-SAT	Vertical order	+.0051	−.0609
Open3DSG	Support/contact	.0000	.0000
Open3DSG	Proximity	+.0968	−.0443
Open3DSG	Vertical order	+.1538	−.3014
SGFN	Support/contact	.0000	.0000
SGFN	Proximity	+.0142	−.0150
SGFN	Vertical order	+.0051	−.0643

Table 17: Family-specific metrics within the selected top-100 predictions. Positive Δ V is a regression; V uses the main verifier.

Source	Condition	V_{all}	V_{dec}	U	V_{pess}
VL-SAT	Source score	.0476	.0729	.3464	.3941
VL-SAT	RelCompat3D	.0295	.0452	.3473	.3768
Open3DSG	Source score	.1242	.2126	.4161	.5402
Open3DSG	RelCompat3D	.0324	.0548	.4085	.4409
SGFN	Source score	.0630	.1005	.3732	.4362
SGFN	RelCompat3D	.0350	.0565	.3803	.4153

Table 18: Uncertainty sensitivity at $K = 100$. Lower is better for all four reported quantities.